Presentation Transcript

Does a Bigger Discount Mean a Better-Rated Product?

Amazon India Product Listings, 2023

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Motivation

Who here has bought a product because it was heavily on sale and you hoped you were getting a great bargain?

Most of us use two signals when shopping online: discount percentage and star rating. Discounts tell us about price, and ratings tell us about satisfaction.

This project asks whether those two signals point in the same direction. If bigger discounts are linked to better ratings, then sale tags are a strong positive signal. If bigger discounts are linked to worse ratings, then consumers should treat deep discounts more cautiously and read reviews more carefully before buying.

So the motivation is simple: use data to make smarter consumer decisions and avoid bargain regret.

Slide 1: Research Question & Dataset

The goal of this study is to answer two related questions. First, does a bigger discount predict a better product rating? Second, do average ratings differ across categories?

The dataset is from Kaggle and contains Amazon India product listings from 2023. After cleaning, I use 1,426 products across three categories:

- Electronics: 526 products
- Computers & Accessories: 453 products
- Home & Kitchen: 447 products

The key variables are discount percentage, average star rating, and category.

The analysis has four parts. I begin with exploratory patterns, then estimate confidence intervals for category means, then run one-way ANOVA and post-hoc pairwise tests, and finally fit OLS regression with controls.

Study design note: what "cross-sectional" means. This dataset is a one-time snapshot: each row is a different product observed in the same period, not the same product tracked over time.

So this is:

- **Cross-sectional** (many products, one time period),
- **Observational** (discounts were not randomly assigned by us).

That is why findings are interpreted as **association, not causation**. We can say higher discounts are linked with lower ratings in this sample, but we cannot claim discounts

themselves cause ratings to fall. Unobserved factors such as product quality, brand reputation, and seller strategy may affect both discounting and ratings simultaneously.

Slide 2: Exploratory Overview

In the exploratory plot, ratings are tightly clustered near 4.0 to 4.3, while discounts vary widely from 0% to 94%.

The first numerical signal is a negative Pearson correlation between discount and rating:

$$r = -0.135, \quad p < 0.001$$

To test whether this correlation is statistically different from zero, I use the standard t-test for correlation:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \qquad \text{df} = n - 2$$

With $n = 1426$ and $r = -0.135$, the test statistic is large in magnitude and negative, so the p-value is below 0.001. This means the negative association is unlikely to be random sampling noise.

At the same time, the size is modest: $r^2 \approx 0.018$, so discount alone explains only about 1.8% of rating variation. So the signal is statistically clear but practically small.

If asked: assumptions behind Pearson correlation

- Independence of observations.
- Approximate linear relationship between variables.
- No extreme outliers dominating the estimate.

If asked: why correlation alone is not enough Correlation is marginal, not conditional. It does not separate discount effects from category composition, which is why we move to ANOVA and then controlled regression.

At this stage, category could still be confounding the relationship, so we need formal group comparisons and controlled regression.

Slide 3: 95% Confidence Intervals for Mean Rating

For each category mean, I use the standard 95% confidence interval:

$$\bar{x} \pm t_{\alpha/2,\, n-1} \cdot \frac{s}{\sqrt{n}}$$

Here, $\bar{x}$ is the sample mean rating, s is the sample standard deviation, and $s/\sqrt{n}$ is the standard error. Because sample sizes are all above 400, these intervals are relatively narrow.

If asked: correct interpretation of a 95% CI If we repeatedly sampled in the same way and built intervals with this rule, about 95% of those intervals would contain the true mean. It is not correct to say there is a 95% probability the fixed mean is inside this one computed interval.

If asked: why use t instead of z Population variance is unknown and estimated from the sample, so the textbook finite-sample interval is t-based. With these large sample sizes, t and z are numerically very close.

The interval results are:

Category	n	Mean	95% CI
Computers&Accessories	453	4.155	[4.131, 4.179]
Electronics	526	4.082	[4.059, 4.105]
Home&Kitchen	447	4.041	[4.010, 4.072]
Overall	1426	4.092	[4.077, 4.107]

The key visual takeaway is that Computers & Accessories and Home & Kitchen do not overlap, which strongly suggests a real difference in their population means.

Electronics overlaps with both, so it sits in between and requires formal hypothesis testing to determine which differences are statistically reliable.

Slide 4: One-Way ANOVA: Rating by Category

ANOVA formally tests whether all three category means are equal.

$$H_0 : \mu_{\text{Elec}} = \mu_{\text{Comp}} = \mu_{\text{Home}}$$

The alternative is that at least one mean differs.

ANOVA partitions variability into between-group and within-group components:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$$

$$MS_{\text{Between}} = \frac{SS_{\text{Between}}}{k-1}, \qquad MS_{\text{Within}} = \frac{SS_{\text{Within}}}{N-k}$$

The F-test statistic is:

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}}$$

If asked: ANOVA assumptions

- Independent observations across products.
- Within-group errors are approximately normal (less critical at large n).
- Homogeneity of variance across groups for classical one-way ANOVA.

Given large and fairly balanced group sizes, ANOVA is generally robust to moderate deviations from normality.

The ANOVA table is:

Source	SS	df	MS	F
Between (Category)	0.629	2	0.314	18.11
Within (Residual)	24.657	1423	0.0173	
Total	25.286	1425		

Since $p < 0.001$, I reject H_0. At least one category mean is different.

The intuition behind this decision is the size of the F value itself. Because

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}} = 18.11,$$

the between-category signal is about 18 times the within-category baseline noise. If all category means were truly equal, we would expect an F value much closer to 1. So a large observed F together with a very small p-value means the null of equal means is not credible for this sample.

If asked: ANOVA effect size

$$\eta^2 = \frac{SS_{\text{Between}}}{SS_{\text{Total}}} = \frac{0.629}{25.286} \approx 0.025$$

So category explains about 2.5% of total rating variance in this one-way decomposition: statistically clear, practically modest.

After ANOVA, I run post-hoc pairwise comparisons to identify exactly which category pairs differ.

For pairwise tests, I use Welch's t-test, which does not assume equal variances:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

with Welch-Satterthwaite degrees of freedom

$$\nu \approx \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\dfrac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \dfrac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

which is why Welch is preferred when variances may differ.

Because I run three pairwise tests, I apply Bonferroni correction to control family-wise error:

$$\alpha_{\text{adj}} = \frac{0.05}{3} = 0.0167$$

So a pair is significant only if $p < 0.0167$.

Pair	Mean Difference	Verdict
Computers&Acc vs Home&Kitchen	+0.114 stars	**Significant**
Electronics vs Computers&Acc	−0.073 stars	**Significant**
Electronics vs Home&Kitchen	+0.041 stars	Not significant

The star differences are differences in average ratings on the 1–5 scale. For example, +0.114 means Computers & Accessories is higher by 0.114 stars on average than Home & Kitchen.

Notice the Electronics versus Home & Kitchen comparison has a positive difference but is not significant after Bonferroni. So we do not claim that difference as statistically reliable in this multiple-testing setup.

If asked: why Bonferroni Bonferroni is simple, transparent, and controls family-wise Type I error strongly. Given only three comparisons, conservativeness is acceptable and easy to explain.

Slide 5: Why We Control for Category

Category is a confound in this dataset. Home & Kitchen has both lower discounts and lower ratings, while Computers & Accessories has higher discounts and higher ratings.

A confound is a third variable that is related to both the predictor (discount) and the outcome (rating), so it can distort the apparent relationship between them if not controlled.

If we regress rating only on discount, category effects get mixed into the discount slope. That is why I use Model M2:

$$\text{Rating}_i = \beta_0 + \beta_1 \cdot \text{Discount\%}_i + \beta_2 \cdot \mathbf{1}[\text{Electronics}_i] + \beta_3 \cdot \mathbf{1}[\text{Home\&Kitchen}_i] + \varepsilon_i$$

What is a predictor vs an estimator?

- **Predictors** are the input variables in the model: Discount%, the Electronics indicator, and the Home&Kitchen indicator.
- **Parameters** $\beta_0, \beta_1, \beta_2, \beta_3$ are unknown population quantities.
- **Estimators** $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$ are computed from the sample and used in the fitted equation.

In matrix notation, the model is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & \text{Discount\%}_1 & \mathbf{1}[\text{Elec}_1] & \mathbf{1}[\text{Home}_1] \\ 1 & \text{Discount\%}_2 & \mathbf{1}[\text{Elec}_2] & \mathbf{1}[\text{Home}_2] \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

and the OLS estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

So each product contributes one row of predictors, OLS aggregates all rows, and returns the estimated coefficients.

This model holds category constant while estimating the discount effect.

The reference category is Computers & Accessories. So the dummy coefficients are interpreted as shifts relative to that baseline, holding discount fixed.

If asked: what coefficient tests are testing

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0$$

using t-statistics of the form

$$t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

with residual degrees of freedom from the fitted model.

Slide 6: Model M2 — What the Coefficients Mean

The fitted M2 equation is:

$$\widehat{\text{Rating}} = 4.295 - 0.00260 \cdot \text{Discount\%} - 0.082 \cdot \mathbf{1}[\text{Electronics}] - 0.150 \cdot \mathbf{1}[\text{Home\&Kitchen}]$$

Interpretation:

- Each additional 1 percentage point of discount is associated with -0.0026 stars, holding category fixed.
- Electronics rates 0.082 stars below Computers & Accessories at the same discount.
- Home & Kitchen rates 0.150 stars below Computers & Accessories at the same discount.

For practical size, comparing a product at 90% discount to one at 10% discount in the same category (a difference of $90 - 10 = 80$ percentage points):

$$80 \times (-0.00260) = -0.208 \text{ stars}$$

So the effect is small but visible.

Slide 7: Model M2 — Significance and Practical Size

All main predictors are statistically significant in this model, and the discount coefficient remains negative after controlling for category.

For the discount slope specifically, you can emphasize both significance and interval interpretation:

$$\hat{\beta}_{\text{discount}} = -0.0026, \quad p < 0.001$$

and the 95% confidence interval lies entirely below zero, so the direction is stable under sampling uncertainty.

Model fit is modest:

$$R^2 = 0.057$$

This means discount and category explain about 5.7% of rating variation, while the remaining 94.3% is due to unobserved drivers such as product quality, brand reputation, and review composition.

If asked: does low R^2 invalidate significance? No. R^2 measures predictive variance explained, while p-values test whether specific coefficients are distinguishable from zero. It is possible to have low R^2 and still estimate a coefficient precisely in large samples.

Slide 8: Model Diagnostics (M2)

Residual diagnostics are critical because OLS inference assumes linearity, constant variance, and approximately normal errors for classical p-values and confidence intervals.

For the Residuals versus Fitted plot, what we want to see is a random cloud centered around zero with roughly constant vertical spread. That is what we largely observe.

What we do not want to see in this plot is:

- A curved pattern, which would indicate nonlinearity and model misspecification.
- A funnel shape, where spread increases or decreases with fitted values, indicating heteroskedasticity.
- Distinct clusters by fitted value that suggest omitted categorical structure.

For the Q-Q plot, what we want is points close to the 45-degree reference line. We see moderate lower-tail deviation from that line, which indicates residual non-normality in the lower tail.

What we do not want in Q-Q diagnostics is:

- Strong S-shaped curvature, suggesting major skewness.
- Heavy tail departures on both ends, suggesting strong outlier influence.
- Systematic departure across the full range, suggesting poor distributional fit.

In this dataset, non-normality is expected because ratings are bounded on $[1, 5]$ and massed near high values. So mild departures do not invalidate the main coefficient-level conclusions.

With $n = 1{,}426$, coefficient inference is generally robust, but prediction intervals at extreme discount values should be treated cautiously.

If asked: heteroskedasticity concern The residual plot does not show a severe funnel pattern. If one wanted an extra conservative check, heteroskedasticity-robust standard errors (HC-type) could be reported as a sensitivity analysis.

Slide 9: Why Are Bigger Discounts Associated with Lower Ratings?

There are at least three plausible explanations for the negative association:

1. Lower-quality products are discounted more aggressively to clear inventory.
2. Deal framing raises expectations, and disappointment leads to lower ratings.
3. Budget brands discount more and also have lower ratings, so discount proxies for quality tier.

This is observational data, so I treat these as plausible mechanisms, not proven causal pathways.

An important interpretation point is that a negative slope does not imply "discounts cause dissatisfaction." A more defensible statement is: conditional on observed category, deeper discounts co-occur with lower ratings in this sample.

Slide 10: Conclusions & Limitations

My three main conclusions are:

1. Category differences are clear and stronger than the discount effect.
2. Bigger discounts are consistently associated with slightly lower ratings.
3. Overall model fit is weak, so most rating variation is driven by unobserved factors.

Key limitations are no causal identification, omitted variables such as brand and product quality, bounded rating outcomes, and a single market-year snapshot.

Data source: Kaggle Amazon Sales Dataset (Amazon India product listings and reviews, 2023).
`https://www.kaggle.com/datasets/karkavelrajaj/amazon-sales-dataset`